

COMPLEX FOURIER SERIES

2l-periodic function which satisfies Dirichlet's conditions can be represented by a real Fourier series:

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \underbrace{a_n \cos \frac{n\pi x}{l} + b_n \sin \frac{n\pi x}{l}}$$

$$a_0 = \frac{1}{l} \int_{-l}^l f(u) du$$

$$a_n = \frac{1}{l} \int_{-l}^l f(u) \cos \frac{n\pi u}{l} du$$

$$b_n = \frac{1}{l} \int_{-l}^l f(u) \sin \frac{n\pi u}{l} du$$

From Euler formula ($e^{i\alpha} = \cos \alpha + i \sin \alpha$)

$$\begin{cases} e^{i\alpha} = \cos \alpha + i \sin \alpha \\ e^{-i\alpha} = \cos \alpha - i \sin \alpha \end{cases}$$

$$\cos \alpha = \frac{e^{i\alpha} + e^{-i\alpha}}{2}$$

$$\sin \alpha = \frac{e^{i\alpha} - e^{-i\alpha}}{2i}$$

$$\cos \frac{n\pi x}{l} = \frac{1}{2} \left[e^{i \frac{n\pi x}{l}} + e^{-i \frac{n\pi x}{l}} \right]$$

$$\sin \frac{n\pi x}{l} = \frac{1}{2i} \left[e^{i \frac{n\pi x}{l}} - e^{-i \frac{n\pi x}{l}} \right]$$

$$\begin{aligned} \underbrace{a_n \cos \frac{n\pi x}{l} + b_n \sin \frac{n\pi x}{l}} &= a_n \frac{1}{2} \left[\underline{e^{i \frac{n\pi x}{l}}} + \underline{e^{-i \frac{n\pi x}{l}}} \right] + \\ &+ b_n \frac{1}{2i} \left[\underline{e^{i \frac{n\pi x}{l}}} - \underline{e^{-i \frac{n\pi x}{l}}} \right] = \\ &= \underbrace{\left(\frac{a_n}{2} - \frac{ib_n}{2} \right)}_{c_n} \underline{e^{i \frac{n\pi x}{l}}} + \underbrace{\left(\frac{a_n}{2} + \frac{ib_n}{2} \right)}_{c_{-n}} \underline{e^{-i \frac{n\pi x}{l}}} = \end{aligned}$$

$$= \sum_{n=-\infty}^{+\infty} c_n e^{i \frac{n\pi x}{l}}$$